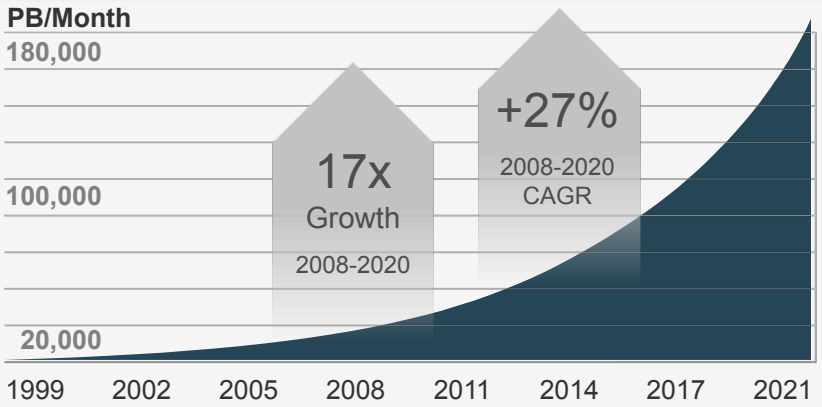
ネットワークの高速化と効率的利用
～インタフェースとASICの高速化～

ジュニパーネットワークス株式会社
サービスプロバイダ事業本部
チーフアーキテクト　長滝信彦

爆発的なトラフィック増加に伴う収益の悪化

1 爆発的なトラフィック増加



Source: Juniper, Cisco, MINTS

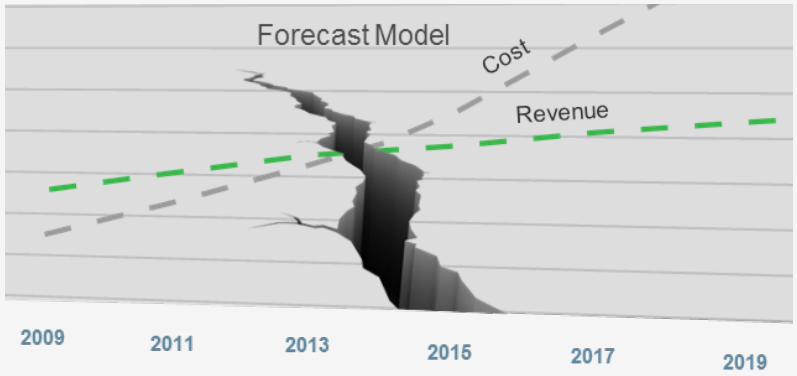
2 ダイナミックなトラフィックパターン

Mobility

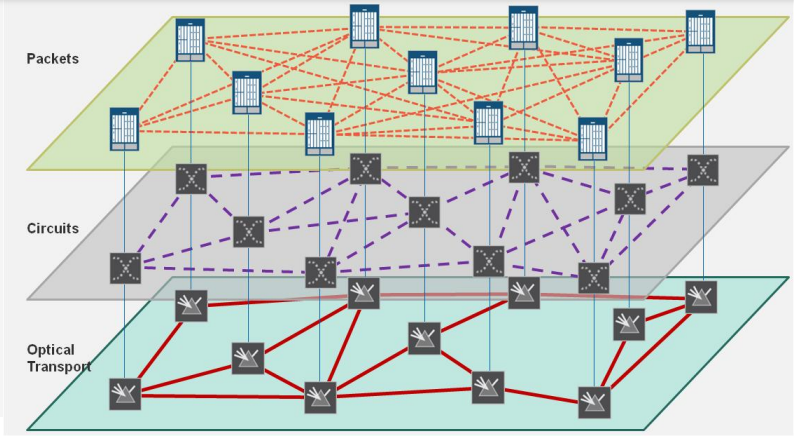
IP Apps

Cloud

3 ビジネスモデルのチャレンジ



4 複雑化したネットワーク

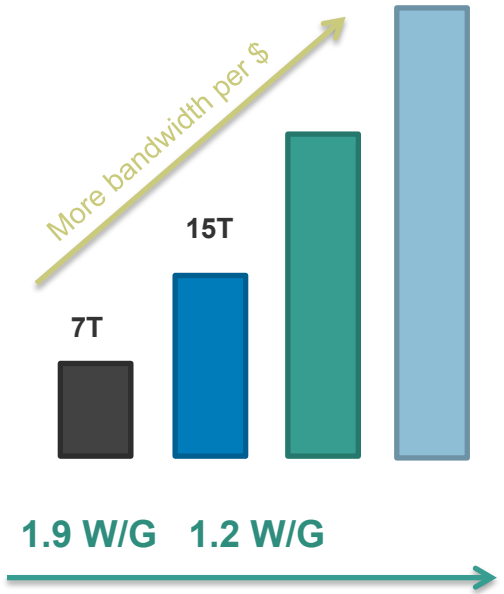


Copyright © 2014 Juniper Networks, Inc.

ルータベンダーとしての問題解決アプローチ

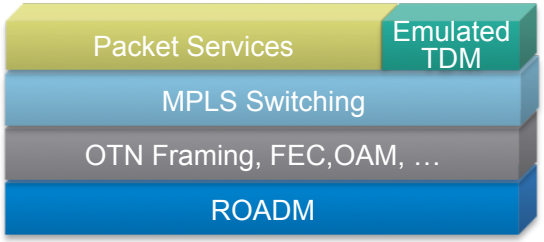
広帯域化

シングルシャーシのスイッチング容量 (Tbps)



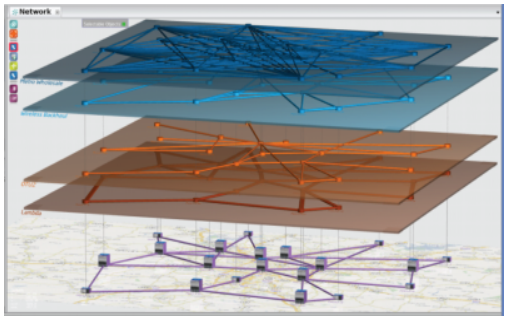
パケット・オプティカル統合

- レイヤ数・装置数削減
- 統計多重効果



SDN & NFV

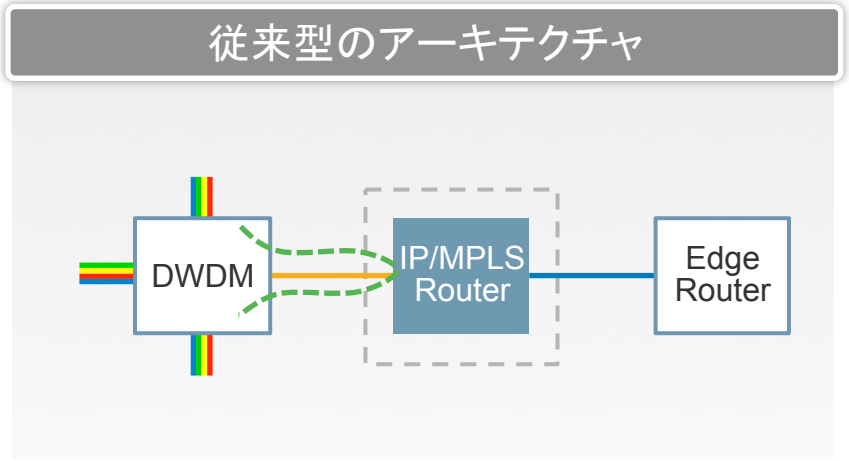
- マルチレイヤでのネットワーク最適化
- PCEなどを用いたLSPの最適化

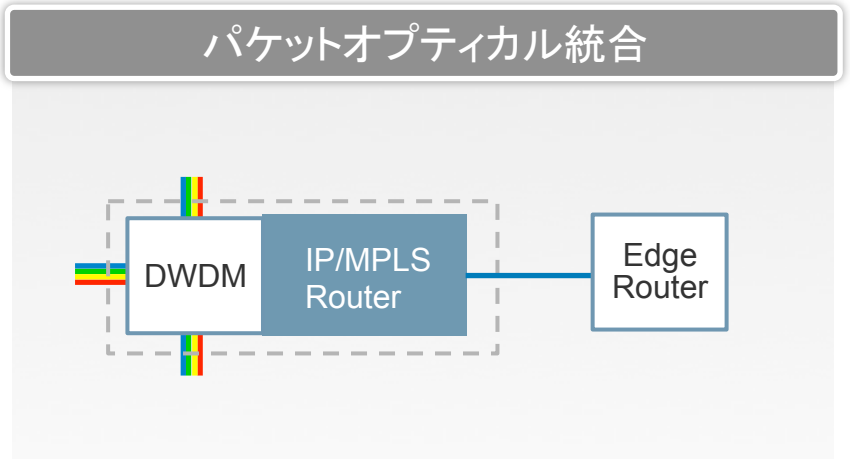


Copyright © 2014 Juniper Networks, Inc.

問題解決へのアプローチ①　パケット･オプティカル統合

ルータベンダーのアプローチ　→　ルータにOpticalインタフェースを実装

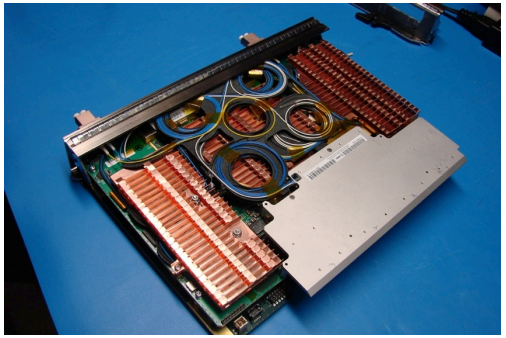




2013年 第一世代DWDMインタフェースモジュールの登場

Parameter	Specification
Wavelength Range	96 channel C-Band 25 GHz ITU-T Grid

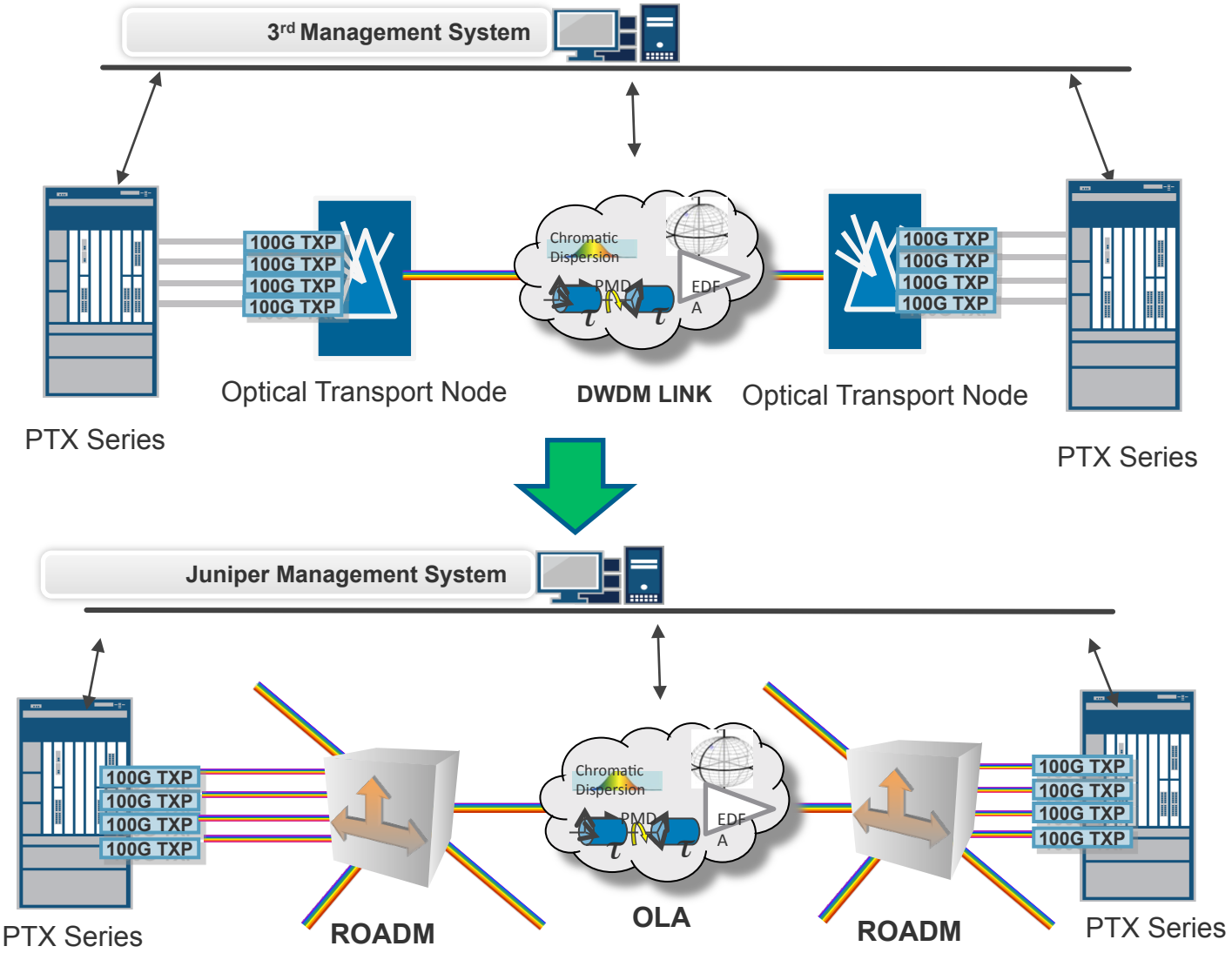
100G DP-QPSK, 2000+ km
Full OTN monitoring



Copyright © 2014 Juniper Networks, Inc.

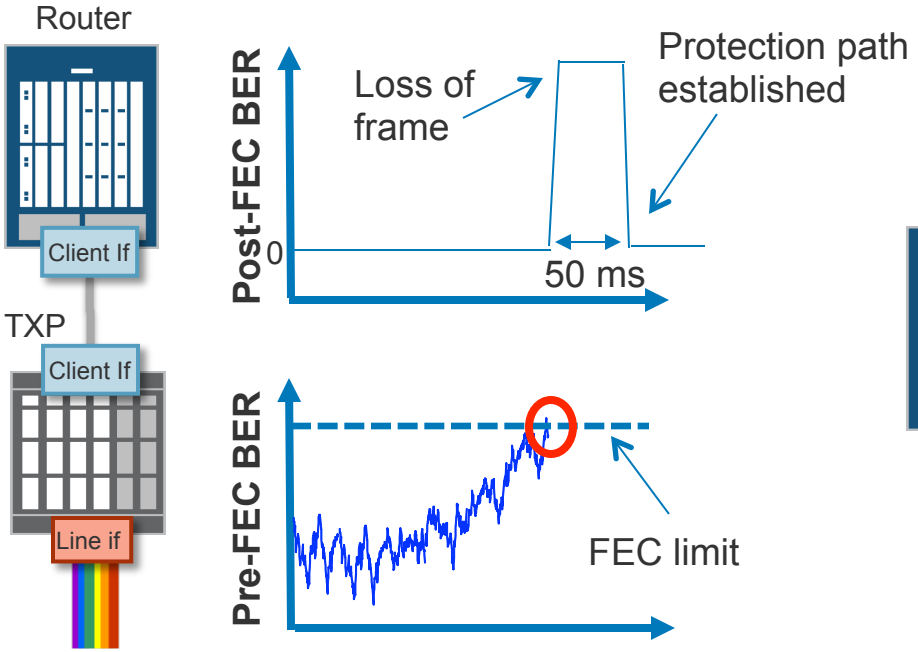
ルータのパケット・オプティカル・ソリューション

- ルータ上に統合100G WDMインタフェース、WDMシステム上にはトランスポンダ無し

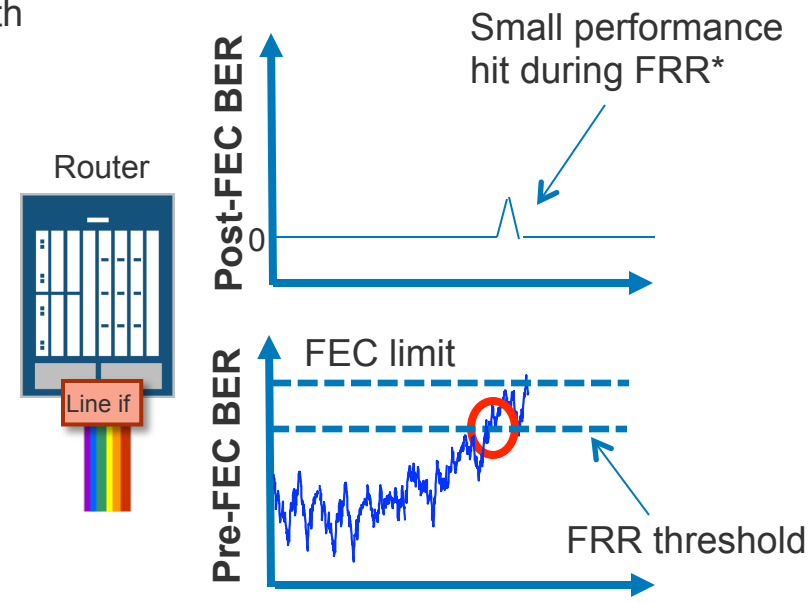


Copyright © 2014 Juniper Networks, Inc. www.juniper.net

パケット・オプティカル統合による利点
PRE-FEC TRIGGERED FRR



Optical layer protection in today's networks



Pre-FEC triggered FRR in converged packet-optical networks

光レイヤでの品質低下をトリガーにFRRを発動することが可能

** Depends on FRR threshold and allocated margin.*

Copyright © 2014 Juniper Networks, Inc.

PRE-FEC TRIGGERED FRR　設定例

```
et-1/0/1 {
optics-options {
        wavelength 1544.53;
    }
    otn-options {
        laser-enable;
        signal-degrade {
            ber-threshold-clear 7;
            ber-threshold-signal-degrade 6;
        }
        preemptive-fast-reroute {
            signal-degrade-monitor-enable;
            backward-frr-enable;
             interval 20;
```





Copyright © 2014 Juniper Networks, Inc.

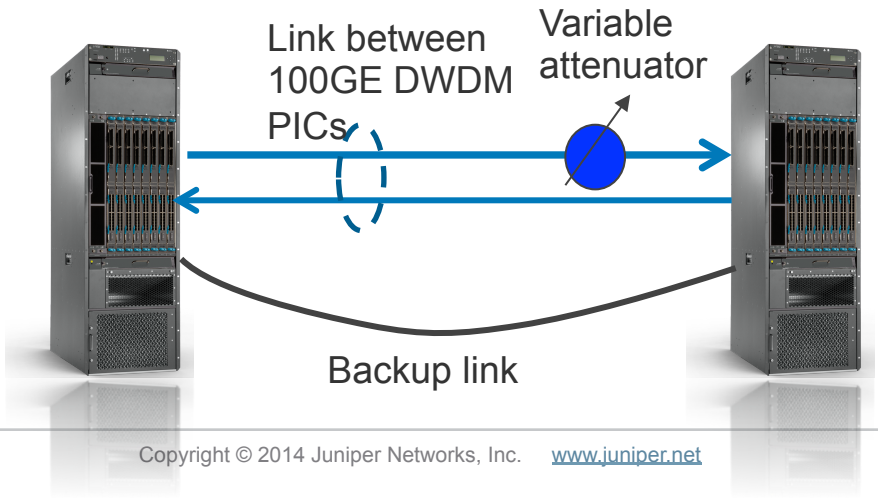
PRE-FEC TRIGGERED FRRのテスト

Without pre-emptive FRR configured:

Test number	Packets lost (Eastbound)	Outage time (Eastbound)	Packets lost (Westbound)	Outage time (Westbound)
1	55247	11.0 ms	56427	11.3 ms

With pre-emptive FRR configured:

Test number	Packets lost (Eastbound)	Outage time (Eastbound)	Packets lost (Westbound)	Outage time (Westbound)
1	0	0	0	0

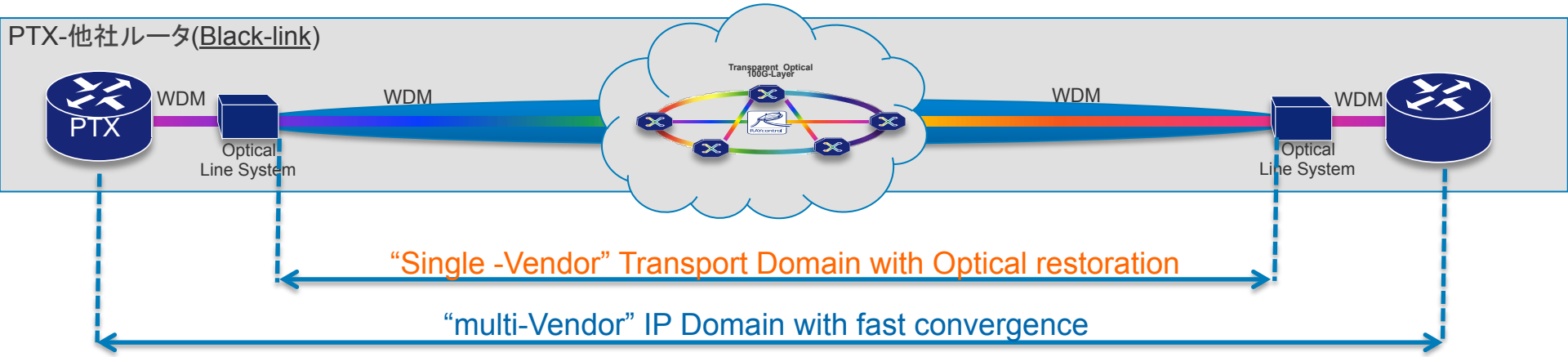


Copyright © 2014 Juniper Networks, Inc. www.juniper.net

ITU-T G.698.2: BLACK LINK
マルチベンダ環境でのインターワーキング

マルチベンダ環境でのインターワーキング
- オプティカルパラメータを標準として定義
  - 最大チャネル数
  - 最大ビットエラー率
  - 最大平均チャネル出力
  - 最小平均チャネル出力

オプティカルパラメータはMIB(Black Link MIB)でも定義し、SNMPで管理



Copyright © 2014 Juniper Networks, Inc.

BLACK LINK 標準化

IETF: Black Link Framework:
http://tools.ietf.org/html/draft-kunze-g-698-2-management-control-framework-01

IETF: Black Link Management:
http://tools.ietf.org/html/draft-galimbe-kunze-g-698-2-snmp-mib-01

ITU-T G.698.2:

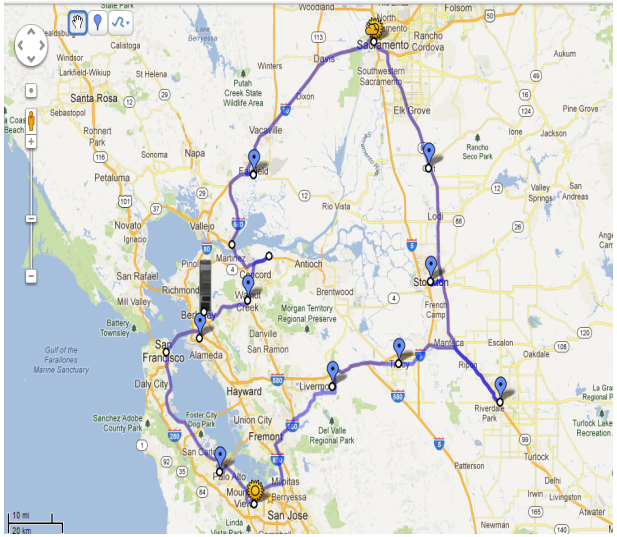
http://www.itu.int/rec/dologin_pub.asp?lang=e&id=T-REC-G.698.2-200911-I!!PDF-E&type=items

Copyright © 2014 Juniper Networks, Inc.

パケット・オプティカル実証実験



Round trip: 1400km w/ 14 ROADMs

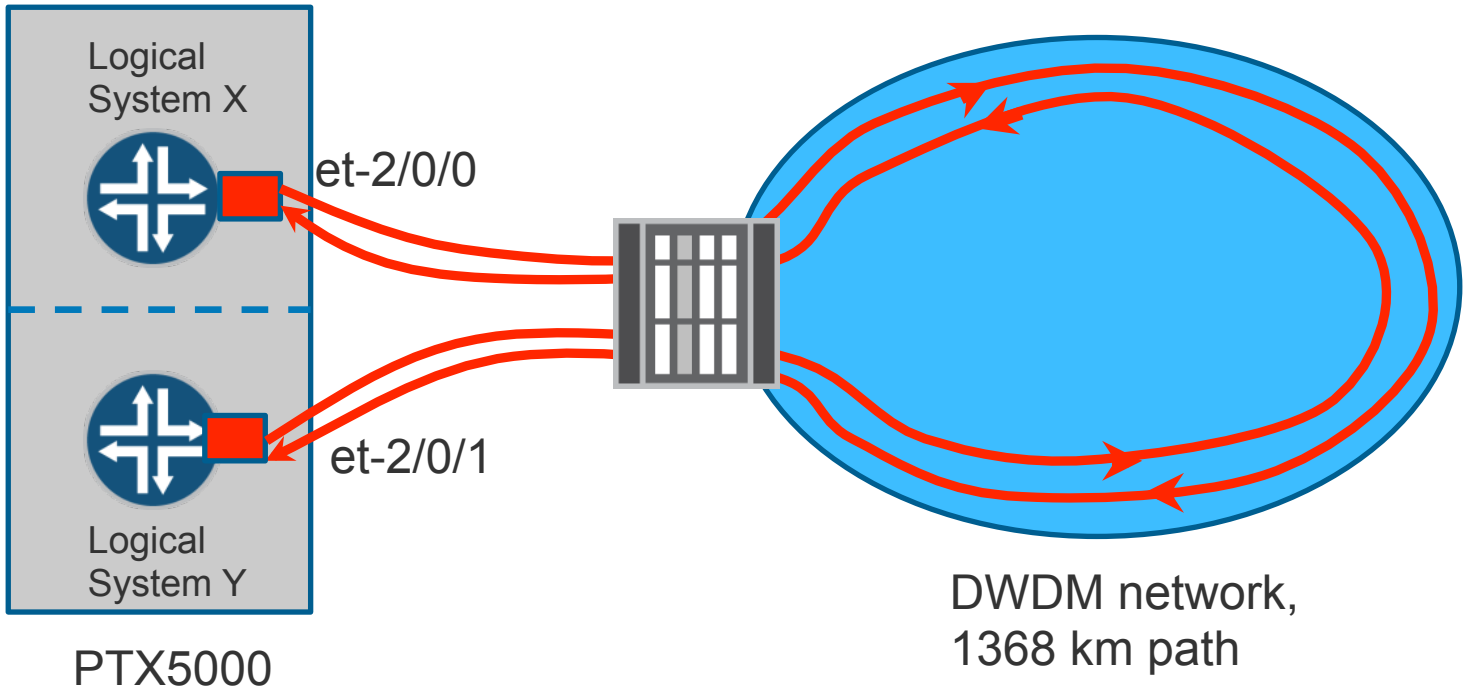


Vertical Simplification: Multilayer Convergence

- 2x100G インタフェースモジュールを使用
- 複数の伝送機器ベンダーとトライアルを実施

Copyright © 2014 Juniper Networks, Inc.

パケット・オプティカル実証実験　接続構成



Copyright © 2014 Juniper Networks, Inc.

パケット・オプティカル実証実験　テスト結果

Interface et-2/0/0:

```
 OTN FEC statistics:
    Corrected Errors                              109579121217328
    Uncorrected Words                                           0
    Corrected Error Ratio (      204252 sec average)   4.22e-03
```

Interface et-2/0/1:

```
  OTN FEC statistics:
    Corrected Errors                              147271954347470
    Uncorrected Words                                           0
    Corrected Error Ratio (      234903 sec average)   4.93e-03
```

- “Show interfaces extensive”の出力結果
- *Uncorrected Words=0* -> post-FECのあとはエラーがない

Copyright © 2014 Juniper Networks, Inc.

パケット・オプティカル実証実験　SNMP MIB

Result Table | Trap Receiver

Name/OID	Value
jnxIfOtnOTUkPreFecBERThresholdMantissa.1.2.1.2	1
jnxIfOtnOTUkPreFecBERThresholdMantissa.1.2.2.2	1
jnxIfOtnOTUkPreFecBERThresholdTime.1.2.1.2	10
jnxIfOtnOTUkPreFecBERThresholdTime.1.2.2.2	10
jnxIfOtnOTUkPreFecBERThresholdExponent.1.2.1.2	6
jnxIfOtnOTUkPreFecBERThresholdExponent.1.2.2.2	6
jnxoptIfOTNPMCurrentFECBERExponent.512.1	10
jnxoptIfOTNPMCurrentFECBERMantissa.512.1	78
jnxRxOpticalPower.1.2.1.2	-1008
jnxRxOpticalPower.1.2.2.2	-2130
jnxoptIfOTNPMIntervalFECCorrectedErr.511.1.1	9832551698
jnxoptIfOTNPMIntervalFECCorrectedErr.511.1.2	10447986451
jnxoptIfOTNPMIntervalFECCorrectedErr.511.1.3	9179953714
jnxoptIfOTNPMIntervalFECCorrectedErr.511.1.4	9198356664
jnxoptIfOTNPMIntervalFECCorrectedErr.511.1.5	11859070334

Pre-FEC BER threshold = 10^{-6}

Pre-FEC BER = 7.8×10^{-9}

Received optical power = -10.08 dBm

Most recent 15 min interval

of corrected errors in this 15 min interval

of corrected errors in this 15 min interval

of corrected errors in this 15 min interval

of corrected errors in this 15 min interval

of corrected errors in this 15 min interval

(…Etc up to jnxoptIfOTNPMIntervalFECCorrectedErr.511.1.96)

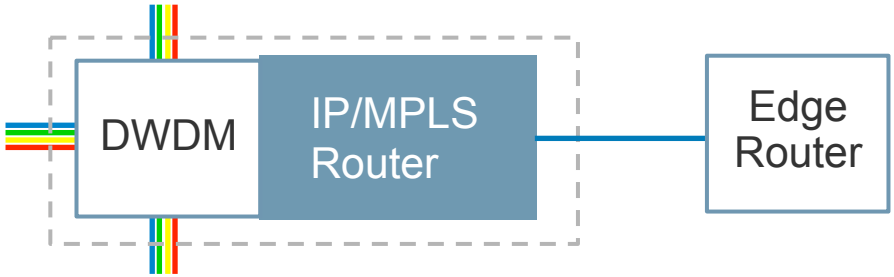
Copyright © 2014 Juniper Networks, Inc.

問題解決へのアプローチ②　インタフェースの広帯域化

ルータのインタフェースの広帯域化で考えなければならないこと。

①100G超のオプティカル

②100G超のイーサネット



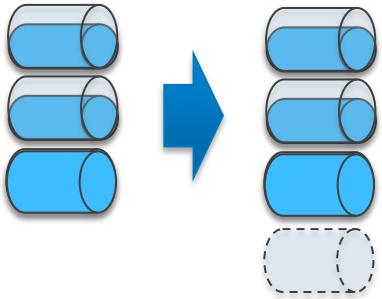
③各コンポーネントの高速化

Copyright © 2014 Juniper Networks, Inc.

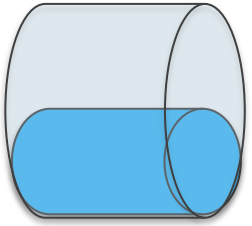
その前に、、
リンクアグリゲーションの課題

一般的な課題
回線本数が増えると
ロードバランスが大変

LAG上の物理ポート

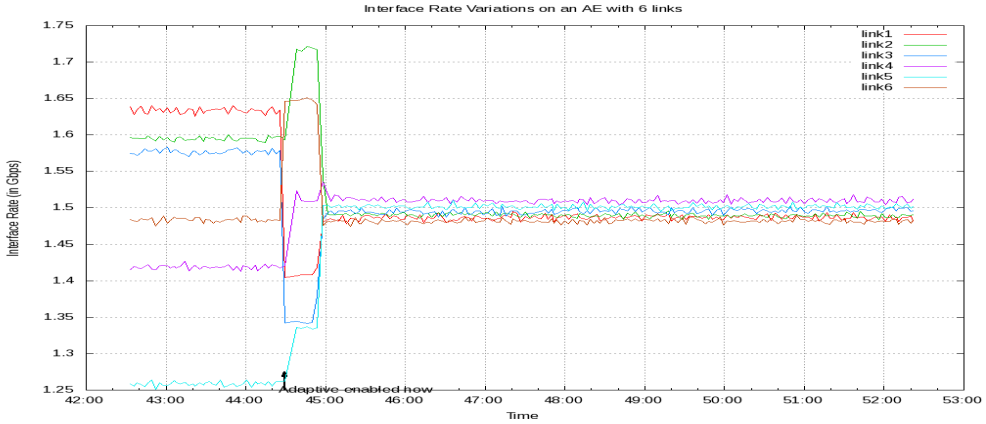


LAG トラフィックトータル で発生し得る状況



Adaptive Load Balancing =LAG内トラフィックの平準化

ALB利用前　**ALB利用後**



ポート数抑制に寄与。
*弊社機能比較

Copyright © 2014 Juniper Networks, Inc.

400G インタフェースに関する動向

IEEE　400GEインタフェース標準化
ITU　　100G超　次世代OTNの標準化

400G DWDMのトライアル



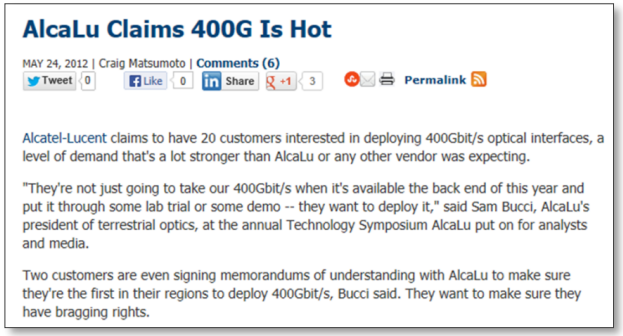
KPN Trials 400G With Huawei

JUNE 22, 2012 | Post a Comment

Tweet 0 Like 0 Share +1 0 Permalink

AMSTERDAM -- Huawei, a leading global information and communications technology (ICT) solutions provider, and KPN International, a leading telecommunications and ICT provider, announced the successful completion of the world's first pan-European 400G field trial. The OTN-based 400G network spans 540 km between Amsterdam, the Netherlands, and Dusseldorf, Germany. There is ample margin in the system, confirmed by adding 300km in the end terminal station. No longer confined to the laboratory, the successful trial of 400G long-haul transmission on a live network marks a major leap forward for 400G in the region.

The exponential growth of traffic on the carrier backbone is accelerating the large-scale commercial deployment of 100G. The Huawei 100G solution stretches over 55,000 km worldwide, with the high-speed transport industry paying increasing attention to the technology beyond 100G with a keen eye on 400G.



AlcaLu Claims 400G Is Hot

MAY 24, 2012 | Craig Matsumoto | Comments (6)

Tweet 0 Like 0 Share +1 3 Permalink

Alcatel-Lucent claims to have 20 customers interested in deploying 400Gbit/s optical interfaces, a level of demand that's a lot stronger than AlcaLu or any other vendor was expecting.

"They're not just going to take our 400Gbit/s when it's available the back end of this year and put it through some lab trial or some demo -- they want to deploy it," said Sam Bucci, AlcaLu's president of terrestrial optics, at the annual Technology Symposium AlcaLu put on for analysts and media.

Two customers are even signing memorandums of understanding with AlcaLu to make sure they're the first in their regions to deploy 400Gbit/s, Bucci said. They want to make sure they have bragging rights.



France Telecom, AlcaLu Deploy 400G Link

FEBRUARY 06, 2013 | Post a Comment

Tweet 5 Like 30 Share +1 0 Permalink

PARIS -- France Telecom-Orange and Alcatel-Lucent (Euronext and NYSE: ALU) today announced the deployment of the world's first optical link offering a capacity of 400 Gigabits per second (Gbps) per wavelength in a live network. Following a successful field trial, a 400 Gbps wavelength optical link is now operational between Paris and Lyon. This link, which was deployed within France Telecom-Orange's operational environment, represents a groundbreaking milestone for long-distance terrestrial network technology.

With capacity four times higher than the maximum bandwidth currently available and using 44 wavelengths, the new optical link can transmit up to 17.6 Terabits per second (Tbps) of traffic in total.



Sprint Picks Ciena for 400G

JUNE 13, 2012 | Craig Matsumoto | Post a Comment

Tweet 0 Like 0 Share +1 0 Permalink

Sprint Nextel Corp. plans to upgrade its optical backbone with 400Gbit/s coherent wavelengths running on Ciena Corp. gear, the vendor is announcing Wednesday.

The new backbone will run on Ciena's 6500 Packet-Optical Platform exclusively, Ciena says. The 400Gbit/s part won't come right away, as the interfaces aren't shipping yet. Sprint and Ciena are planning for field trials to run in early 2013.

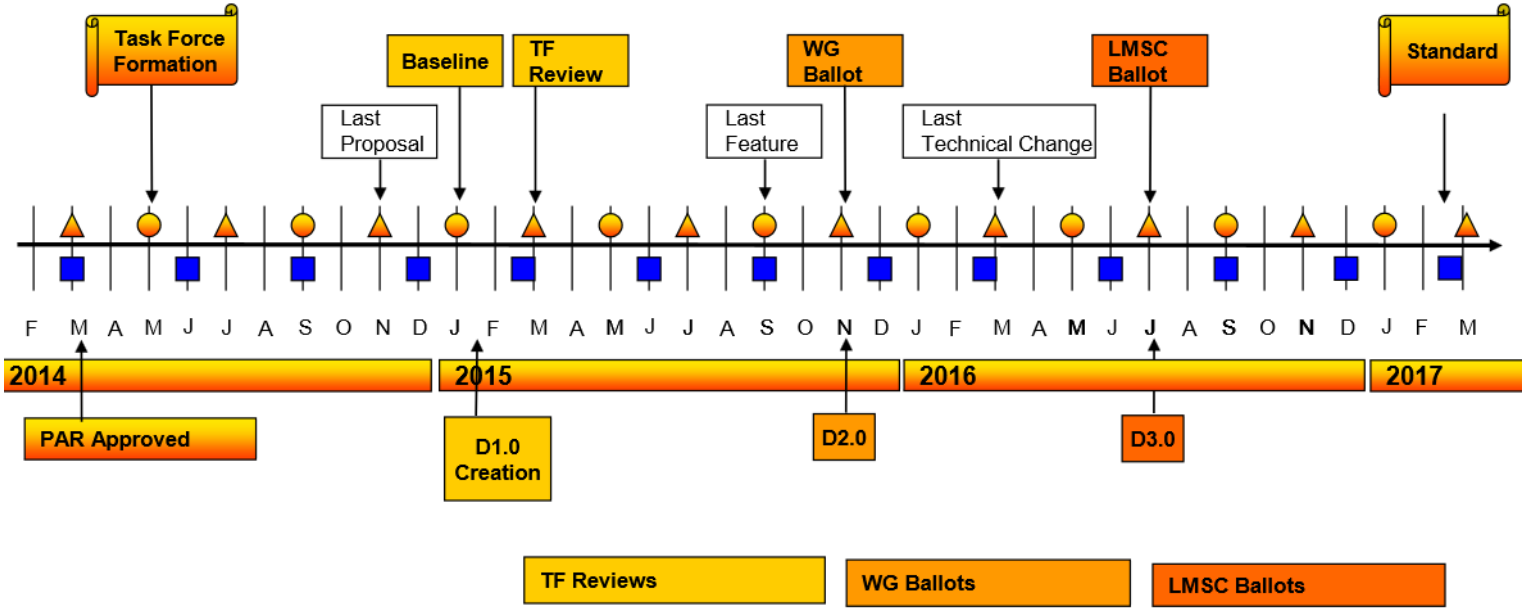
Sprint was already using Ciena's optical gear in general. The upgraded backbone is part of Sprint's overall Network Vision plan, which includes Alcatel-Lucent, Ericsson AB and Samsung Corp. in other capacities. (See What I Learned at Sprint....)

Ciena has already said that BT Group plc will be a 400Gbit/s customer as well.

Copyright © 2014 Juniper Networks, Inc.

IEEE 400GE標準化の動向

- 2012年9月　　IEEE voted to standardize 400G
- 2013年4月　　IEEE 802.3 400 Gb/s Ethernet Study Group (http://www.ieee802.org/3/400GSG/).
- 2014年5月　　IEEE P802.3bs 400Gb/s Ethernet Task Force
- 2017年　　標準化完了見込み



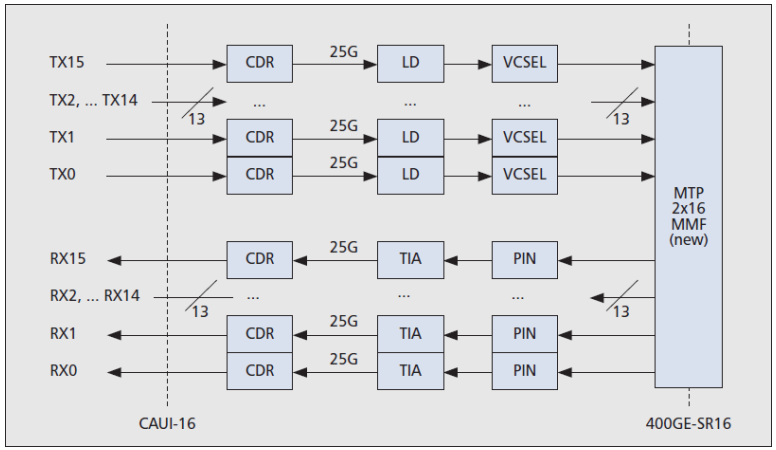
Source: http://www.ieee802.org/3/bs/public/14_05/dambrosia_3bs_01a_0514.pdf

Copyright © 2014 Juniper Networks, Inc.

400G クライアントインタフェース

マルチモード400GEインタフェース

- SR16（16 x 25G parallel multi-mode optics）が議論されている
  - 100G SR4 standard proposal (see IEEE 802.3bm) と非常によくにた方式
  - 100m程度の短距離接続にフォーカス



Source: Finisar

シングルモード400GEインタフェース

- 複数の方式が議論されている
  - Eg. Fiber-ribbon SMF for datacenter interconnects (“PSM4”) using single-mode MPO cabling

Copyright © 2014 Juniper Networks, Inc. www.juniper.net

CDFP (400GBPS FORM-FACTOR PLUGGABLE) MSA

400Gクライアントインタフェースのトランシーバーモジュール

- データセンターがメインのアプリケーション
  - 25Gbps x 16レーンに対応する高密度のpluggable form factor
- 2014年5月に、機械的な仕様を発表

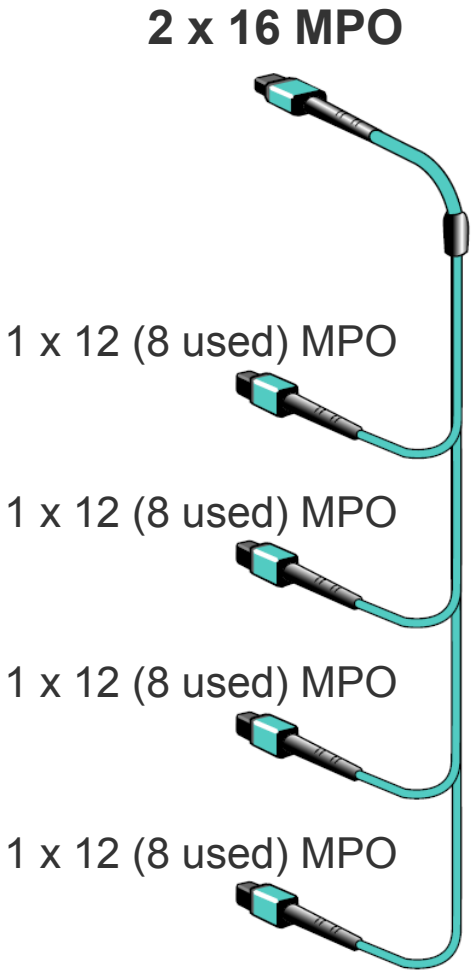
New Multi-Source Agreement (MSA) Launching to Advance 400 Gbps Copper Cable and Fiber Optic Transceiver Markets

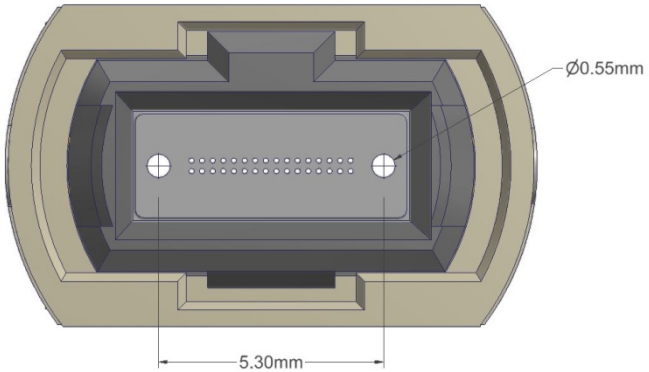
CDFP MSA forming to develop standard transceiver specifications for 400 Gbps data transmission rates

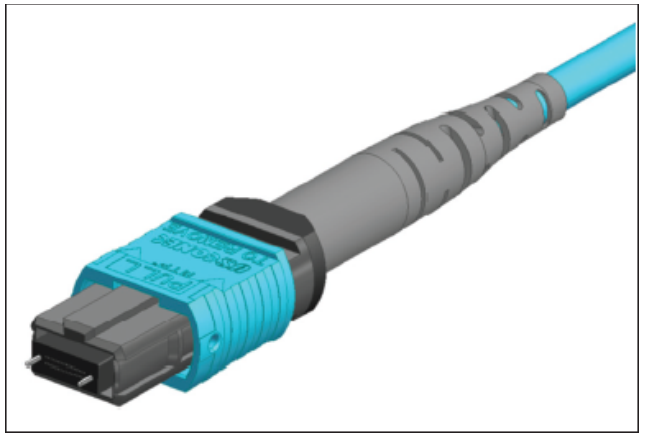
ANAHEIM, CA– March 19, 2013 – Five leading global companies intend to enter into a multi-source agreement (MSA) to create the CDFP (400 Gbps form-factor pluggable) industry consortium to define a transceiver module/plug mechanical form factor and a host-board electrical edge connector and cage.

Copyright © 2014 Juniper Networks, Inc.

MMF BREAK-OUT CABLES: FROM 100G TO 400G







Parallel Multi-Mode with 16 lanes:

- 4 x 100GBASE-SR4
- 16 Tx Fibers and 16 Rx Fibers
- 2x16 MPO connector (multi-mode)

100G から 400G へのマイグレーション用途を想定

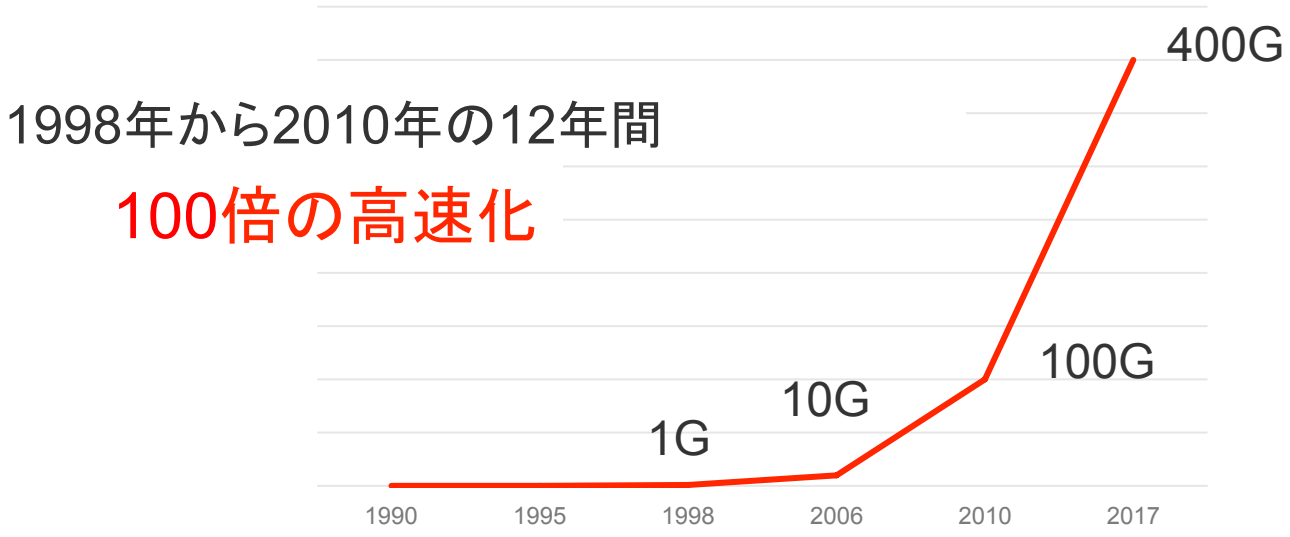
Copyright © 2014 Juniper Networks, Inc. www.juniper.net

イーサネット高速化技術の歴史

1998年 IEEE 802.3z (1000Base-X) 標準化

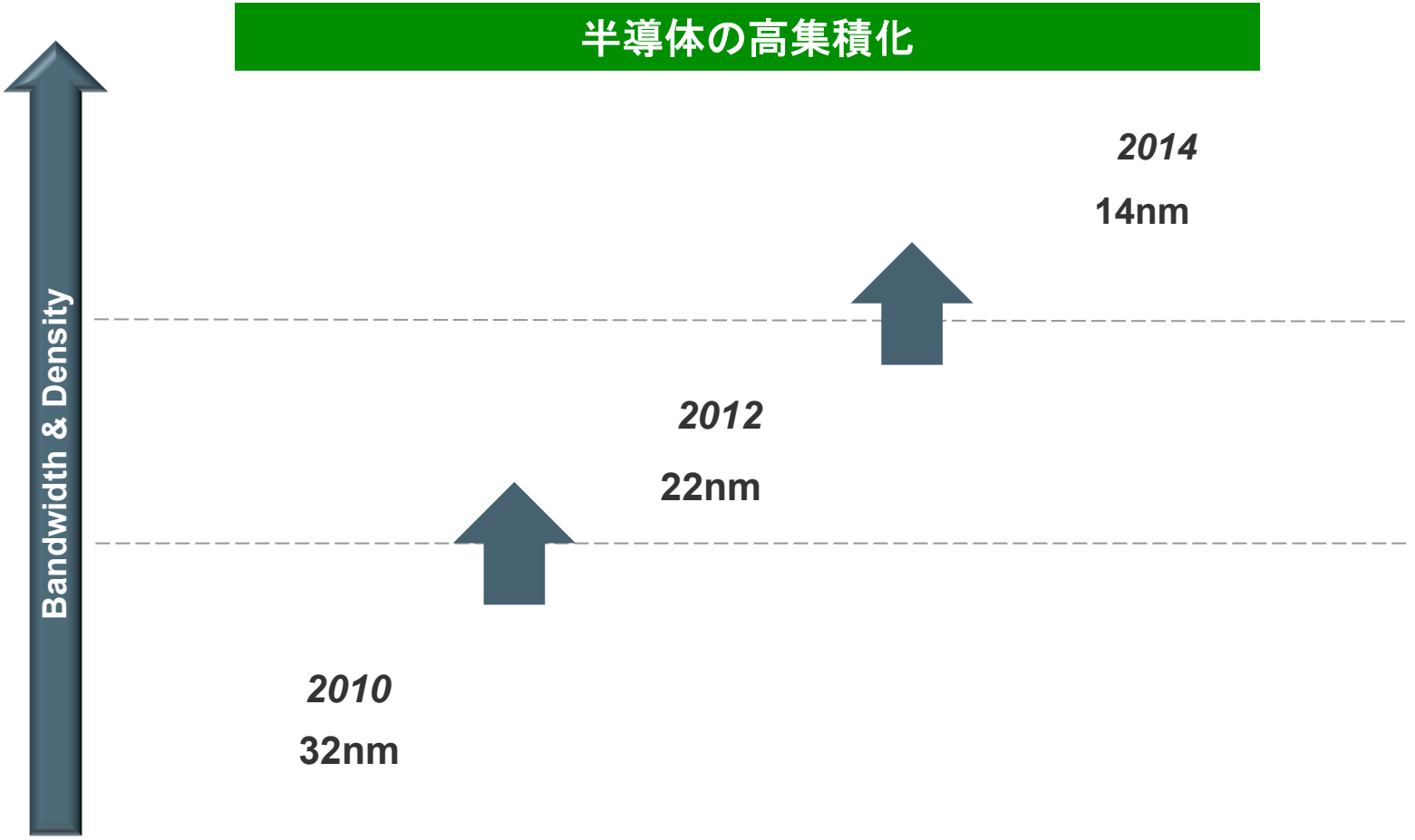
2006年 IEEE 802.3ae (10GBase-xR) 標準化

2010年 IEEE 802.3ba (100GBase-xR) 標準化



Copyright © 2014 Juniper Networks, Inc.

半導体の高集積化



Copyright © 2014 Juniper Networks, Inc. www.juniper.net

まとめ

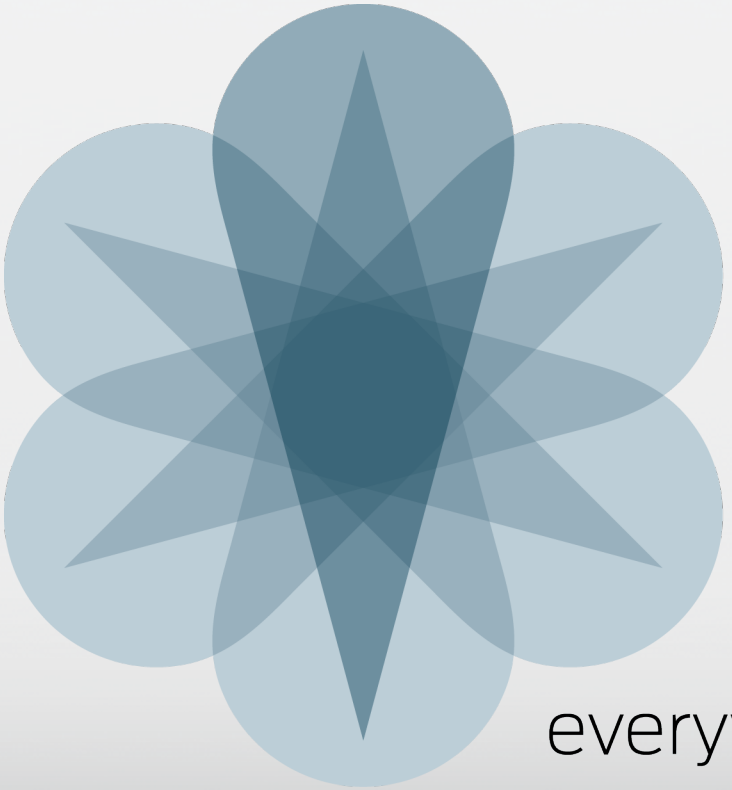
爆発的なトラフィック増加に対応するために、、

広帯域化　パケット・オプティカル統合　SDN & NFV

①高速インタフェースの標準化
400GE、次世代OTNの策定

②各コンポーネントの開発
最新のオプティカル、半導体技術の導入

Copyright © 2014 Juniper Networks, Inc.



everywhere